OpenHIE Auth & Bring-up Setup

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Local / single-host bring-up. Steps and commands only. Run from the repo root. `<like-this>` = placeholder.

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0. Prerequisites

Install these first:

- **Docker Engine + Compose v2** — https://docs.docker.com/engine/install/ (verify: `docker compose version`)
- **Git** — https://git-scm.com/downloads
- CLI tools: `jq`, `curl`, `openssl`, `python3`, `wg` / `wg-quick` (Debian/Ubuntu: `sudo apt-get install -y jq curl openssl python3 wireguard`)
- WireGuard config for the JeMPI VPS (`10.8.0.1`, peer `202.51.71.81`)

Then get the code:

```
git clone https://github.com/ETech-Nepal/openhim.git openhie
cd openhie
for d in mediator admin-mediator patient-portal-mediator fhir-intake-mediator; do
  cp $d/.env.example $d/.env
done
```

The examples are preconfigured for the compose network. The only value you must edit for bring-up is `OPENHIM_PASSWORD` in all four — done in §2c, once you have the password from §2b.

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1. Containers

```
docker compose up -d
docker compose ps
```

Running: `hapi-db`, `mongo-db`, `openhim-core`, `openhim-console`, `fhir`, `keycloak`, `mediator`, `admin-mediator`, `patient-portal-mediator`, `fhir-intake-mediator`.

> **Never `docker compose down`** (wipes Keycloak's H2). Use `stop` / `start`. Mediators crash-loop until steps 2 and 5 are done.

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2. OpenHIM first login

2a. Accept the self-signed cert

1. Open https://localhost:8080/heartbeat, click through the warning.

2. CLI needs `-k`: `curl -k https://localhost:8080/heartbeat`

2b. Root password

1. Console http://localhost:9000, log in `root@openhim.org` / `openhim-password`.
2. Set a new password. Record it.

2c. Propagate to mediators

Set `OPENHIM_PASSWORD` in `mediator/.env`, `admin-mediator/.env`, `patient-portal-mediator/.env`, `fhir-intake-mediator/.env`, then:

```
docker compose up -d --force-recreate mediator admin-mediator patient-portal-mediator
        fhir-intake-mediator
```

> `restart` does **not** reload `.env` — use `up -d --force-recreate`.

Console → **Mediators**: all four green.

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3. Reach JeMPI

The mediator talks to JeMPI at `JEMPI_BASE_URL` in `mediator/.env`. Pick one. After changing it: `docker compose up -d --force-recreate mediator`.

Option A — JeMPI public IP (simplest)

```
# mediator/.env
JEMPI_BASE_URL=http://202.51.71.81:50000/JeMPI
```

Check: `curl http://202.51.71.81:50000/JeMPI/countRecords`

No tunnel. Traffic crosses the public internet unencrypted (JeMPI has no TLS/auth of its own) — test data only; switch to Option B for real patient data.

Option B — WireGuard tunnel

Keeps the default `JEMPI_BASE_URL=http://10.8.0.1:50000/JeMPI`.

```
sudo wg-quick up wg0                                    # config: /etc/wireguard/wg0.conf
curl http://10.8.0.1:50000/JeMPI/countRecords           # from host
docker exec mediator wget -qO- http://10.8.0.1:50000/JeMPI/countRecords
```

Host works but the container hangs — masquerade the docker subnet out `wg0`:

```
sudo iptables -t nat -A POSTROUTING -s 172.16.0.0/12 -o wg0 -j MASQUERADE
sudo sysctl -w net.ipv4.ip_forward=1
```

`wg0.conf` `AllowedIPs` must include `10.8.0.0/24`.

> You can also run the tunnel **and** point `JEMPI_BASE_URL` at the public IP `http://202.51.71.81:50000/JeMPI` — both reach the same JeMPI.

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4. Keycloak realm

Admin Console http://localhost:8083, bootstrap admin `admin` / `admin`.

```
curl -fsS http://localhost:8083/realms/openhie/.well-known/openid-configuration |
        jq .issuer
```

If it prints `http://localhost:8083/realms/openhie`, the realm exists — go to §6. Otherwise:

- Realm dropdown → **Create realm** → name `openhie` → **Create**
- Or `./keycloak/setup.sh`

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5. Realm key into OpenHIM

```
./keycloak/extract-realm-pubkey.sh
docker compose up -d --force-recreate openhim-core
docker compose restart mediator admin-mediator patient-portal-mediator fhir-intake-
        mediator
```

> A realm rebuild rotates Keycloak's RSA key → re-run this or every token fails `JWT could not be verified`.

OpenHIM env (already in `docker-compose.yml`): `authentication_jwt_algorithms` = `RS256` (plain string), `authentication_jwt_audience` = `account`, `authentication_jwt_issuer` = `http://localhost:8083/realms/openhie`.

Transaction-port errors:

Code	Meaning
401	token missing / bad signature / issuer or audience mismatch
403	Client mapped, but lacks a channel role
404	no channel matched the URL

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6. Per-hospital client

One command per hospital — NHFR code + name. Creates the Organization, keypair, Keycloak client + `sub`-override mapper, and OpenHIM Client; verifies the token `sub`:

```
./keycloak/onboard-hospital.sh 3130400221 "Bharatpur Hospital"
```

Idempotent. Output gives the private-key path — hand it to the hospital, delete your copy.

System	Value
FHIR	`Organization.identifier` = `<nhfr-id>`; `Organization.id` = server UUID
Keycloak	client `<nhfr-id>`, `private_key_jwt`, token `sub` = `<nhfr-id>`
OpenHIM	Client **ID** = `<nhfr-id>`, role `np-fhir-clients`
Mediator	`X-OpenHIM-ClientID` = `<nhfr-id>` → `GET Organization?identifier=nhfr|<nhfr-id>`

Prereqs: §1, §2c, §5.

Just an Organization, no client: `./keycloak/create-org.sh <nhfr-id> "<name>"`.

> NHFR system string: deployed stack uses `http://mohp.gov.np/fhir/NamingSystem/nhfr` (NP Core's canonical is `https://nhfr.mohp.gov.np`).

> `sub` override rejected → `onboard-hospital.sh` fails at 4/5. Fallback: OpenHIM `clientID` stays the Keycloak UUID; add `FACILITY_REGISTRY_JSON={"<uuid>":"<nhfr-id>"}` to `mediator/.env` + `docker compose up -d --force-recreate mediator`.

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7. Smoke test

```
TOKEN=$(./keycloak/get-jwt-client-token.sh 3130400221)

curl -k -sS -X POST https://localhost:5000/fhir/Patient \
  -H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/fhir+json' \
  -d '{"resourceType":"Patient","name":[{"family":"SHRESTHA","given":["KAMANA"]}],
       "gender":"female","birthDate":"2001-06-25",
       "telecom":[{"system":"phone","value":"9800000000"}]}'
```

`201` with a UUID `id` and `nepal-health-id` in `identifier` = the full chain works. Console → **Transactions** → the request → **orchestrations**.

Postman: `node keycloak/token-server.mjs`, then `http://localhost:4099/token/3130400221`.

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8. FHIR Intake mediator (`/ingest`)

`POST /ingest` in front of the orchestrator:

`Content-Type`	Body	Action
`application/json`	one raw CDRS encounter row	validate → Patient + Encounter Bundle → forward
`application/fhir+json`	resource or `transaction` / `batch` Bundle	forward unmutated
other	—	`415`

Channel `NP FHIR Intake`, `^/ingest/?$`, `POST`, route `fhir-intake-mediator:4003`, role `OPENHIM_CHANNEL_ROLE` in `fhir-intake-mediator/.env` (`np-fhir-clients` to reuse §6 clients).

Raw-row behaviour: `403` if no `X-OpenHIM-ClientID`; `GET Organization?identifier=nhfr|<nhfr-id>` → `422` if 0 or >1 match; stamps `managingOrganization` / `serviceProvider` = the resolved `Organization/<uuid>`. Inbound `X-NHFR-ID` / `organization_id` ignored.

8a. Dump script

```
cd FHIR_mediator
python3 -m venv venv && venv/bin/pip install -r requirements.txt
cp .env.example .env          # KEYCLOAK_CLIENT_ID=<nhfr-id>, PRIVATE_KEY_FILE,
        SQLITE_PATH

DRY_RUN=1 venv/bin/python main.py      # preview
venv/bin/python main.py                # raw rows
venv/bin/python main.py --FHIR         # client-mapped Bundles
venv/bin/python main.py --all          # also patients with no insurance number
```

Progress bookmarked per `encounter_number`.

8b. Smoke test

```
TOKEN=$(./keycloak/get-jwt-client-token.sh 3130400221)

curl -k -sS -X POST https://localhost:5000/ingest \
  -H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
  -d '{"hospital_patient_number":"2405078","encounter_number":"SMOKE-1",
       "first_name":"CHANDRIKA","last_name":"KANDEL","sex":"Female",
       "department_type":"C","department_id":"22","department_name":"Family Medicine",
       "visit_date_ad_time":"2026-07-28 00:02:00.000"}'
```

`201`. A second POST of the same `encounter_number` must not create a duplicate.

Quick reference

Task	Command
Create the realm	`./keycloak/setup.sh`
Create/update an Organization	`./keycloak/create-org.sh <nhfr-id> "<name>"`
Re-extract realm key	`./keycloak/extract-realm-pubkey.sh`
Onboard a hospital	`./keycloak/onboard-hospital.sh <nhfr-id> "<name>"`
Hospital token	`./keycloak/get-jwt-client-token.sh <nhfr-id>`
Postman token helper	`node keycloak/token-server.mjs` (`:4099`)
JeMPI tunnel up	`sudo wg-quick up wg0`
Push data	`cd FHIR_mediator && venv/bin/python main.py` (`--FHIR`, `--all`)

Troubleshooting

Symptom	Fix
Console calls fail, network error	accept the cert — https://localhost:8080/heartbeat (2a)
Mediator missing under **Mediators**	`OPENHIM_PASSWORD` ≠ root password (2c)
`.env` change ignored	used `restart`, not `up -d --force-recreate`

Symptom	Fix
`JWT could not be verified`	`realm-public.pem` stale — re-run §5
401 at `:5000` fresh token	issuer mismatch — `iss` must be `http://localhost:8083/realms/openhie`
403 at `:5000`	Client lacks role — re-run `onboard-hospital.sh` (§6)
404 at `:5000/ingest`	`NP FHIR Intake` channel missing — check `fhir-intake-mediator`
403 at `:5000/ingest`	set `OPENHIM_CHANNEL_ROLE=np-fhir-clients` on `fhir-intake-mediator`
422 at `:5000/ingest`	no Organization with NHFR identifier `<nhfr-id>` — `create-org.sh` (§6)
`onboard-hospital.sh` fails at 4/5	Keycloak won't override `sub` — `FACILITY_REGISTRY_JSON` fallback (§6)
container can't reach `10.8.0.1`, host can	masquerade the docker subnet out `wg0` (§3)
Keycloak realm gone after a stop	`docker compose down` was used — use `stop` / `start`